

Exosomes in neurodegenerative diseases: Therapeutic potential and modification methods

Neural Regeneration Research

Introduction

Diseases of the central nervous system impact nearly one-sixth of the global population, with their complex pathogenesis often leading to disability and mortality (Zhou et al., 2021; Helgudóttir et al., 2024; Kia et al., 2024). Among these, neurodegenerative diseases (NDs) exemplify conditions that severely affect central nervous system structures (Liu et al., 2024; Yang et al., 2024). Common examples of NDs include Alzheimer's disease (AD), Parkinson's disease (PD), stroke, amyotrophic lateral sclerosis (ALS), Huntington's disease (HD), multiple sclerosis (MS), and epilepsy. Characterized by intricate etiologies, NDs constitute a group of challenging disorders with a lack of effective treatments, thus imposing substantial burdens on medical systems, society, and the economy (Li et al., 2023). Although various plausible pathogenic mechanisms have been reported, the exact etiological causes of NDs remain elusive (Qi et al., 2024). The pathogenesis of NDs is associated with disruptions in autophagy and lysosomal function, mitochondrial dysfunction, oxidative stress, alterations in protein degradation pathways, the accumulation of misfolded proteins, and neuroinflammatory responses (Rey et al., 2022). Maintaining normal protein homeostasis is crucial for preserving cellular functional integrity; thus, abnormalities in protein homeostasis have been proposed as potential triggers for the glial-cell-mediated inflammatory responses that ultimately lead to ND pathogenesis (Sonninen et al., 2020).

Neurodegeneration, caused by defective synaptic transmission and neuronal failure within the brain, is prevalent among patients with NDs and the elderly and is often accompanied by autoinflammation. This inflammatory environment promotes the activation of microglia and astrocytes in the central nervous system, which helps protect the brain from microbial invasion or injury. Microglia and astrocytes are the primary mediators of neuroinflammation in the brain, and their chronic activation can disrupt tissue homeostasis, resulting in neurotoxic effects (Ransohoff, 2016). Central nervous system infections stimulate localized immune responses by increasing cytokine levels, and sustained infection-induced inflammation triggers the activation of local immune cells. However, the chronic activation of microglia and astrocytes can be detrimental, as it enhances neuroinflammation and accelerates neurodegeneration (Gaire and Choi, 2021). Persistent or uncontrolled neuroinflammation promotes the unregulated secretion of pro-inflammatory cytokines, hindering neuronal restoration and exacerbating neurodegenerative processes (Liu et al., 2022c). One study confirmed the involvement of signal transduction pathways, including type 1 transmembrane glycoproteins, mitogen-activated protein kinases, and nuclear factor kappa B, in amplifying microglia-mediated neuroinflammation (Ye et al., 2022). Understanding the underlying molecular mechanisms of glial cell activation is critical for mitigating neuronal damage, which is a key focus in ND treatments. As the primary immunocompetent cells in the central nervous system, microglia play a pivotal role in the pathological evolution of NDs such as AD and PD, where their morphology and transcriptional functions are altered (West et al., 2022). When steady-state microglia are exposed to proteins or molecules associated with NDs, they transition from a quiescent state to a phagocytic state, changing from a ramified morphology to an amoeboid configuration (Gemma and Bachstetter, 2013). Activated microglia provide protective effects through the phagocytic clearance of exosome-associated misfolded proteins and other harmful agents.

Most cells in the central nervous system, including neural stem cells, microglia, neurons, oligodendrocytes, and astrocytes, can release exosomes, facilitating intercellular material transfer and information exchange in neurological disorders (Lai and Breakefield, 2012). Exosomes from various stem cells mediate processes associated with cell proliferation, apoptosis, angiogenesis, and inflammation, and thereby promote tissue repair and regeneration (Kirkham et al., 2022). Consequently, stem cell-derived exosomes show significant potential for therapeutic applications in tissue regeneration (Vizoso et al., 2017). For instance, mesenchymal stem cell-derived exosomes (MSC-exosomes) exhibit a

range of immunomodulatory, anti-inflammatory, anti-fibrotic, antioxidative, and pro-angiogenic properties. They serve as effective vehicles for targeted drug delivery to pathological sites, offering a platform for precision therapy. A combination of bone marrow stem cell (BMSC)-derived exosomes with conductive hydrogels can promote the regeneration of myelinated axons, thereby improving muscle denervation atrophy (Yang et al., 2023). Exosomes from BMSCs confer neuroprotective effects and alleviate pyroptosis in neurons by inducing favorable microglial polarization (Liu et al., 2021). Notably, human adipose MSC-exosomes selectively infiltrate macrophages and microglia, limiting the cells' activation during brain injury, which mitigates neuroinflammation and facilitates functional recovery (Chen et al., 2020b). Exosomes that bind to biosynthetic cellulose membranes can promote peripheral nerve regeneration and are increasingly used to repair nerve defects (Cui et al., 2023). However, a significant limitation of natural exosomes is their lack of specific cell-recognition capabilities. This deficiency can lead to side effects and toxicity when the exosomes interact with non-specific cells. While exosome-based therapies for NDs are a promising and innovative approach, the modification of exosomes is essential to enhance their therapeutic effects.

In this article, we aim to elucidate the therapeutic potential of exosomes in NDs and explore modification strategies to improve exosome targeting. We review the pathogenesis of various NDs, including AD, PD, stroke, ALS, HD, MS, and epilepsy, and discuss the roles of exosomes in the treatment of these conditions. This review concludes with a discussion of the approaches used to enhance exosome targeting, highlighting their potential as new therapeutic options for NDs.

Search Strategy

The articles cited in this review were obtained from a PubMed database search of studies published from inception to 2025 using the following keywords: stem cell exosomes, central nervous system diseases, neurodegenerative diseases, exosome targeting, Alzheimer's disease, blood–brain barrier, cell recognition, cognitive function, Parkinson's disease, stroke, amyotrophic lateral sclerosis, multiple sclerosis, Huntington's disease, epilepsy, exosome modification, stem cell therapy, and enhanced targeting. The results were filtered based on the titles and abstracts of the articles. Articles that did not discuss the therapeutic role of exosomes in NDs were excluded from consideration. The references cited in this review were screened for full articles written in English. This review includes a total of 199 references, with 114 published in 2020 and beyond. The seminal literature is presented in **[Figure 1]**.

Timeline showing the development and application of exosomes. AD: Alzheimer's disease; EV: extracellular vesicle; MSCs-exo: mesenchymal stem cells derived exosomes; MVBs: multivesicular bodies; PD: Parkinson's disease; TBI: traumatic brain injury.

Exosomes and Neurodegenerative Diseases

Extracellular vesicles comprise a heterogeneous array of submicron-scale particles, including nanoscale exosomes, and have recently garnered significant attention in the field of NDs (Gratpain et al., 2021). These vesicles can be categorized into exosomes (30–100 nm), microvesicles (100–1000 nm), and apoptotic bodies (1000–5000 nm) based on their origin, size, and mechanisms of release (van der Pol et al., 2012). Exosomes can be derived from various cellular sources and are found in bodily fluids such as urine, blood, saliva, and amniotic fluid (Li et al., 2021a). These microscopic vesicles contain a variety of bioactive components, including proteins, lipids, nucleic acids, and other metabolites, both within their interior and on their surface (He et al., 2018). Both normal and cancerous cells generate exosomes, and variations in the specific exosome characteristics and metabolic states of their origin can be analyzed through changes in their composition and content (de Pedro et al., 2013; Koritzinsky et al., 2017). Thus, exosomes may be a potential treatment for NDs.

Exosomes are generated through the endocytic pathway during endosome maturation, which is initiated by the inward growth of the cell membrane to form internal vesicles (Pegtel and Gould, 2019). This maturation process is regulated by

ubiquitin and involves the complex endosomal sorting machinery essential for trafficking. The resulting multivesicular bodies bind to the plasma membrane and release exosomes into the extracellular space (Li et al., 2018; Zhang et al., 2020; [Figure 2]). The membranes of exosomes typically contain ceramides, diacylglycerol, cholesterol, various transmembrane proteins, lysosome-associated membrane glycoproteins, integrins, and transferrin receptors (Kalluri and LeBleu, 2020). As vehicles for drug and gene delivery, exosomes exhibit potential applicability in tissue regeneration, tumor progression, and immune modulation (Zhang et al., 2015a). Additionally, exosomes play crucial roles in coagulation processes, intercellular communication, and cellular waste management (Boriachek et al., 2018). Due to their excellent biocompatibility, stability, low toxicity, and ability to evade the immune system, exosomes can freely cross the blood–brain barrier (BBB) (Wiklander et al., 2015). Their biocompatibility and stability enable them to deliver drugs to target cells reliably, while their low toxicity minimizes the risk of immune responses being triggered. Furthermore, exosomes can deliver therapeutic agents to diseased areas that are difficult to reach with traditional therapies. Exosomes also exhibit cell-tropic properties, allowing for targeted drug delivery, particularly of biological agents such as proteins or nucleic acids (e.g., short interfering RNA or microRNA [miRNA]) (Kojima et al., 2018). Their function is generally realized through the fusion of their surface membrane proteins with the membranes of target cells, which facilitates the transfer of nucleic acids, proteins, lipids, and other components to recipient cells. This interaction can lead to corresponding functional changes in the recipient cells. Moreover, the different routes of drug delivery for exosomes greatly influence their biological distribution and therapeutic effects. A previous review summarized the different administration routes of exosomes in detail (Xu et al., 2021), and the main administration methods for the treatment of NDs include intravenous, oral, stereotactic, and nasal routes. Numerous investigations have explored the therapeutic applications of exosomes in various NDs, including AD, PD, stroke, ALS, HD, MS, and epilepsy.

Composition and biogenesis of exosomes. Following endocytosis, the endocytic material enters early endosomes, where the membrane buds inward to form internal vesicles, initiating endosomal maturation. MVBs are degraded through fusion with lysosomes or release exosomes by fusing with the plasma membrane. MVB: Multivesicular body.

Exosomes and Alzheimer's disease

AD is an irreversible ND and the most common form of dementia and primarily affects individuals in old or pre-old age, thus the risk of developing AD increases with age (Li and Wang, 2025). The disease impacts both memory and cognitive abilities, and has been referred to as the “eraser for the brain.” Its etiology involves a combination of genetic, environmental, immunological, and lifestyle factors (Serrano-Pozo et al., 2021). The pathogenesis of AD is believed to involve the irregular deposition of amyloid- β ($A\beta$), the aggregation of hyperphosphorylated tau protein within cells leading to the formation of neurofibrillary tangles, and neuroinflammatory responses (Khan et al., 2022). Less widely accepted hypotheses include the involvement of mitochondrial dysfunction and synaptic transmission impairment. The abnormal accumulation of $A\beta$ both inside and outside nerve cells can result in neurocommunication disorders and specific neuronal destruction. In AD, $A\beta_{1-40}$ -induced damage to BV-2 microglia disrupts the balance of energy metabolism and neuroinflammation, increasing the release of pro-inflammatory cytokines. Current therapeutic approaches for AD primarily involve cognitive enhancers aimed at improving cognitive function and delaying disease progression (Loera-Valencia et al., 2019). However, existing clinical drugs for AD only alleviate specific symptoms and do not halt the progression of the disease (Kabir et al., 2020). Notably, microglia can mitigate AD progression through the phagocytosis of $A\beta$, thereby inhibiting the activation of certain pro-inflammatory pathways, improving neuroinflammation, and preventing cognitive impairment (Wang et al., 2022a). Despite considerable research efforts, translating effective AD treatments from the laboratory setting to clinical application remains a significant challenge, underscoring the urgent need for novel treatment strategies.

Exosomes have the ability to cross the BBB and may delay the pathological process by promoting the clearance of $A\beta$ or tau. Certain studies have also highlighted their potential as biomarkers for the early detection of AD (Gao et al., 2021;

Duan et al., 2024). For instance, on isolating exosome-enriched extracellular vesicles from the brains of tau transgenic rTg4510 and control mice, Polanco et al. (2016) discovered that these vesicles carried tau with varying levels and phosphorylation states. Their research contributed to advancing the diagnosis of tau-induced AD pathology and provided new insights into potential treatment avenues. Research on AD treatment via exosomes has particularly focused on the therapeutic use of stem cell-derived exosomes ([Table 1] and [Figure 3]). MSC-exosomes are proficient at crossing the BBB and selectively binding to target cells, making them ideal carriers of genetic material for the central nervous system. Li et al. (2024a) found that neural stem cell-derived exosomes promoted the restoration of aberrant protein distribution in APP/PS1 mouse models. Additionally, MSC-exosomes loaded with PTEN-targeting miR-223 activate the PI3K/Akt pathway and reduce neuronal apoptosis, providing a potential approach for treating AD (Wei et al., 2020). Exosomes offer a promising avenue for more efficaciously influencing the progression of AD by traversing the BBB, introducing a novel diagnostic and therapeutic strategy for this condition.

Summary of various exosomes in AD therapeutics

Overview of the role of exosomes in Alzheimer's disease. (A) Pathophysiology: Exosomes released by brain cells cross the BBB and enter the peripheral circulatory system. (B) Treatment: Exosomes acting in conjunction with drugs or that are intravenously injected directly into mice can cross the BBB and target the damaged area to exert a therapeutic effect. BBB: Blood-brain barrier.

Exosomes and Parkinson's disease

PD is the most common ND after AD, and its incidence increases exponentially with age. The average age of onset for PD is slightly older than that of AD, and there are sex differences, with women exhibiting a higher rate of levodopa-induced dyskinesia (Xiaoxia et al., 2022). The typical symptoms of PD fall into two categories: motor symptoms (including bradykinesia, resting tremors, and postural instability) and non-motor symptoms (which encompass psychiatric, sensory, and sleep disturbances). Early symptoms can be elusive, leading to misdiagnosis, while advanced PD significantly incapacitates patients, imposing substantial burdens on both individuals and their families (Berrío Sánchez et al., 2019). The etiology of PD is linked to a complex interplay of factors, including aging, genetics, and environmental influences. The primary pathological feature is the death of dopaminergic neurons in the substantia nigra (Cacabelos, 2017). As previously discussed, neuroinflammation plays a crucial role in safeguarding the brain; however, chronic neuroinflammation can lead to neuronal damage (Kip and Parr-Brownlie, 2022). Notably, the increased expression levels of pro-inflammatory factors in the brain have emerged as inflammatory biomarkers for the detection of PD (Liu et al., 2022c). Treatments for PD include medication and surgical interventions. However, current therapeutic strategies for PD remain imperfect (Goulding et al., 2020). While dopamine pharmacotherapy is effective in the early stages, its prolonged use can result in diminished efficacy and adverse effects (Asahina et al., 2013). Deep brain stimulation—often referred to as brain pacemaking—is applied in advanced cases and involves the implantation of electrodes in deep brain regions to inhibit aberrant neural signals and alleviate symptoms. However, it does not address axial symptoms such as postural gait disturbances. Emerging research has indicated that music, dancing, and physical activities can help alleviate symptoms and reduce the risk of developing PD (Pereira et al., 2019; Xu et al., 2019). Nevertheless, these approaches only provide partial symptomatic relief and do not alter the disease's progression, and there remains a need for novel therapeutic strategies.

Cell replacement therapy has emerged as a potential straightforward remedy for numerous conditions. Given their differentiation potential, MSCs have been explored for use in such cell replacement treatments. However, they struggle to efficiently traverse the BBB without the aid of penetrants, presenting a challenge for their application in stem cell therapy. As significant paracrine agents of MSCs, exosomes can easily cross the BBB, making them a potentially more effective treatment for PD. One study reported that injecting MSC-exosomes into the substantia nigra and striatum of a rat model of PD improved the animals' exercise capacity and histological symptoms (Teixeira et al., 2017). Additionally,

miRNAs can be transferred to neuronal cells via exosome loading. A notable example is miR-133b, an miRNA that promotes neurite outgrowth and is specifically expressed in dopaminergic neurons but is absent in the brains of patients with PD (Kim et al., 2007). BMSC exosomes containing glioma-associated oncogene homolog 1 have been shown to reduce inflammatory damage and neuronal apoptosis *in vitro* and in mouse models of PD by inhibiting Sp1 signaling (Cai et al., 2022). This study provided a basis for the potential clinical application of exosomes derived from BMSCs in the treatment of PD. Furthermore, transgenic macrophages release exosomes containing antioxidants, catalase, and glial-cell-derived neurotrophic factor, which have demonstrated efficacy in ameliorating the symptoms of PD (Zhao et al., 2014). The therapeutic potential of exosomes has been established in various PD models ([Table 2](#)), highlighting exosomes as a promising avenue for treatment.

Summary of various exosomes in PD treatment

Exosomes and stroke

Globally, stroke is associated with high disability and mortality rates, and it significantly impairs the normal functioning of various brain regions. The primary symptoms of stroke include an uneven mouth, unclear speech, and blurred vision. The ischemic and hemorrhagic forms of stroke represent major health concerns, accounting for 75%–90% and 10%–25% of cases, respectively. The ischemic cascade triggers the intracellular influx of sodium and calcium, generating inflammatory cytokines and free radicals, which lead to cell edema and activation of the apoptotic pathway (Dhir et al., 2020). The etiology of stroke is complex and involves factors such as hypertension, diabetes, hyperlipidemia, unhealthy lifestyle habits (e.g., smoking and poor sleep), and obesity. Treatment varies from person to person, with surgical options depending on the type of stroke, onset of symptoms, and detailed vascular anatomy. Currently, no specific drugs have been approved for stroke treatment; instead, only symptomatic supportive therapies are available. These include neuroprotective drugs (e.g., edaravone) and agents that improve cerebrovascular circulation (e.g., aspirin). Regulating neuroinflammation is a promising approach to treating ischemic stroke. The neuroinflammation that occurs in the acute stage of stroke is linked to BBB rupture and neuronal injury, with the mechanisms of inflammatory injury in stroke primarily involving oxidative stress, the increased production of matrix metalloproteinases, and microglial activation (Eduardo et al., 2022). Astrocytes and microglia rapidly activate during a stroke episode, producing excessive reactive oxygen species via mitochondrial oxidase and the NADPH pathway. This oxidative stress harms neurons and glial cells, contributing to neuroinflammation and BBB disruption (Zhu et al., 2022). Intravenous recombinant tissue plasminogen activator can quickly dissolve any blood clots that are obstructing arteries, making it an effective treatment for acute stroke (Emberson et al., 2014). However, its administration window is limited to 4.5 hours post-stroke, and administration beyond this timeframe carries risks that may outweigh benefits (Lees et al., 2010). For example, hemorrhagic transformation and oxidative stress are significant clinical risks that could exacerbate the pathological cascade (Peña et al., 2017). Given the limitations of existing stroke treatment strategies, the pursuit of novel therapies is crucial.

Stem cell therapy has emerged as an effective method to reduce brain and peripheral inflammation following a stroke (Anthony et al., 2022). For instance, Yew et al. (2021) injected human pulp stem cells into the infarcted and adjacent brain tissues of Sprague–Dawley rats and reported improved mobility and the reduced activation of astrocytes. The effects were potentially linked to substances secreted by stem cells, including exosomes. The therapeutic potential of stem cells is likely mediated by paracrine factors contained in the exosomes released by endosomes. Moreover, MSC-exosomes were shown to contribute to nerve recovery after stroke and reduce cerebral infarction volume (Dehghani et al., 2021). Exosomes carry functional agents, such as miRNAs, that influence neurovascular units and are critical in the pathogenesis and recovery of stroke (Rahmani et al., 2020). The intravenous injection of MSC-exosomes in ischemic stroke models has been shown to enhance neurological recovery, promote angiogenesis and neurogenesis, and reduce the expression of pro-inflammatory factors (Xu et al., 2020). Similarly, the intravenous injection of human urine-derived

stem cell exosomes alleviated the neurological deficits and encouraged neurogenesis after ischemic stroke in rats (Ling et al., 2020). Zhang et al. (2019) targeted exosome-mediated miR-210 in ischemic brains to promote angiogenesis following cerebral ischemia, presenting a viable strategy for ischemic stroke treatment. Post-stroke, the exosomes synthesized by brain cells are released through the BBB, and their presence in the peripheral blood or cerebrospinal fluid can serve as biomarkers that reflect the pathological progression and promote recovery following stroke (Otero-Ortega et al., 2019). Studies have demonstrated that stem cell exosomes contribute to the pathological treatment of stroke and improve neuroinflammation ([Table 3]), highlighting exosomes as a novel and potent treatment strategy.

Summary of various exosomes in stroke treatment

Exosomes and other neurological diseases

In addition to the previously discussed NDs, other common conditions, including ALS, HD, MS, and epilepsy, could be treated with exosome therapy to improve outcomes ([Table 4]).

Summary of various exosomes in other ND therapeutics

ALS is a chronic, progressive neurological disease that causes motor neuron and limb damage. The precise cause of ALS remains unknown; however, it is associated with various factors, including genetics, lifestyle, metabolic disorders, and autoimmune abnormalities. The disease primarily affects individuals over the age of 45, with a higher incidence in men than women. The primary symptoms of ALS include skeletal muscle weakness, muscle atrophy, and muscle fasciculations, with symptoms gradually worsening over time. Some individuals with ALS may also experience breathing difficulties, limb paralysis, arrhythmia, and mood disorders such as depression and anxiety. Ultimately, respiratory muscle weakness and failure frequently lead to patient death. Currently, the diagnosis of ALS relies on clinical manifestations and electromyography, and there are no targeted treatment drugs or cures available. Existing treatments, including respiratory support, psychotherapy, and medications, can only improve quality of life and slightly delay disease progression.

A hallmark of ALS are the aggregations of TAR DNA-binding protein 43 (Kim et al., 2020). Exosomes have the potential to clear these aggregates by encapsulating TAR DNA-binding protein 43 (Iguchi et al., 2016), and thus represent a novel treatment strategy for ALS. Schwann cells are crucial for motor neuron function, and their impairment has been observed in ALS. A study demonstrated the safety of utilizing exosomes derived from allogeneic Schwann cells in treating Schwann cell damage in patients with ALS (Goldschmidt-Clermont et al., 2025). Furthermore, exosomes found in plasma may serve as biomarkers for the molecular diagnosis of ALS and for monitoring disease progression (Chatterjee et al., 2024). Exosomes derived from human bone marrow endothelial progenitor cells have been proposed as a new therapeutic approach to repair the compromised blood–central nervous system barrier in ALS (Sadanandan et al., 2021). Another study analyzed the miRNA expression profiles of relevant exosomes in ALS patients and compared them to those of healthy individuals, finding that miR-34a was significantly dysregulated in ALS patients. This suggests that the exosome-mediated overexpression of specific miRNAs may be beneficial for ALS treatment (Rizzuti et al., 2022). Additionally, exosomes derived from adipose mesenchymal stem cells can modulate the microglial phenotype in ALS by reducing reactive oxygen species levels and promoting differentiation toward an anti-inflammatory phenotype (Dabrowska et al., 2024). Bonafede et al. (2020) found that exosomes derived from adipose mesenchymal stem cells reduced spinal motor neuron loss in superoxide dismutase 1 mice, decreased glial cell activation, and improved motor symptoms. Their study highlighted the benefits of using exosomes in nerve repair for ALS.

HD, also known as major chorea or Huntington's chorea, is an autosomal dominant ND that typically presents between the ages of 30 and 45. It is characterized by a chronic, insidious onset of dance-like movements, behavioral abnormalities, and progressive dementia. This incurable disorder is caused by the repeated amplification of CAG trinucleotide repeats in the gene encoding the huntingtin protein. The fragmentation and gradual accumulation of

mutated huntingtin proteins disrupt cellular function, ultimately leading to neuronal cell death (Ananbeh et al., 2021). As with many NDs, specific treatments for HD are limited. Current approaches primarily focus on symptomatic treatment to delay disease progression, improve patients' quality of life, and prevent complications. Recent research suggests that exosomes may offer a novel therapeutic avenue for HD. For instance, astrocyte-derived exosomes were shown to effectively inhibit the aggregation of mutated huntingtin proteins in HD mice (Hong et al., 2017). Additionally, miRNA modulation plays a crucial role in the pathology and recovery of HD, with exosome-carrying miRNAs—particularly miR-128a and miR-196a—demonstrating potential benefits in HD models (Tung et al., 2021).

MS is characterized by demyelination and is the most prevalent chronic inflammatory disease affecting the function of the central nervous system (McGinley et al., 2021). It is a debilitating condition that impacts crucial body structures such as the brain and spinal cord. MS is more common in young and middle-aged women, and it often manifests as paralysis of the eyes, muscles, and limbs. The exact cause of MS remains unknown; however, it is associated with a combination of genetic factors, environmental triggers such as viral infections, and geographical influences that may lead to autoimmune reactions. While a cure for MS remains elusive, current treatment options include medication, infusion therapy, and rehabilitation training. Recently, exosomes have emerged as a potential therapeutic modality for this condition. The activation of the purinergic receptor P2X7 has been implicated as harmful in autoimmune diseases. Research demonstrated increased concentrations of purinergic receptor P2X7 in oligodendrocyte-derived exosomes from MS patients compared with healthy individuals (Agliardi et al., 2024); thus, the levels of exosome-associated purinergic receptor P2X7 may serve as a biomarker for this disease. The exosome-based targeted delivery of bryostatin 1 has been shown to significantly enhance remyelination and provide neuroprotection in animal models of demyelinating MS (Wu et al., 2022). Additionally, Zheng et al. (2023) demonstrated that the intranasal administration of exosomes loaded with resveratrol in MS mice significantly inhibited inflammatory responses in both their central and peripheral nervous systems. Human interferon- γ exosomes contain various anti-inflammatory and neuroprotective proteins that have been shown to significantly reduce demyelination and inhibit inflammation in experimental autoimmune encephalomyelitis (EAE) models (Riazifar et al., 2019). EAE models, which closely resemble the pathology of MS, are widely used for studying the disease (Li et al., 2019a).

Epilepsy is caused by the abnormal discharge of neurons in the brain and is characterized by repeated and transient seizure episodes. The molecular pathways involved in the seizures primarily include Ca^{2+} influx into neurons, coupled with an excitotoxicity induced by glutamate (Zhang and Bhavnani, 2006). Various factors can trigger epilepsy, including muscle contractions, brain tumors, central nervous system infections, and genetic predispositions. The onset of epilepsy can occur at any age, thus the condition affects both the elderly and the young. While the symptoms of epilepsy can vary, they tend to be consistent between episodes for each individual. Currently, there is no specific treatment plan for epilepsy; however, existing therapies primarily include drug treatments to control the condition and reduce the frequency of seizures, aiming to bring the patient's quality of life as close to normal as possible. Globally, approximately 500,000 individuals are adversely affected by epilepsy, many of whom do not respond to standard treatments (Cano et al., 2023). Therefore, there is an urgent need for new therapeutic options. Exosomes have shown promise as potential treatments for epilepsy. For example, when intranasally administered to mice with lipopolysaccharide-induced epilepsy, exosomes derived from BMSCs were found to mitigate microglial activation, suppress hippocampal neuroinflammation, and promote normal neurogenesis (Long et al., 2017).

Exosome therapy is a cell-free approach that offers several advantages, including the ability to efficiently cross the BBB, elimination of the need for immunosuppression, and the capacity to incorporate anti-inflammatory agents. In the field of NDs, exosome therapy has been shown to reduce neuroinflammation, prevent pathological protein aggregation, and alleviate neuropathy ([Figure 4]). While preclinical studies on the use of exosomes in treating NDs have shown promising results, further efforts are needed to facilitate their widespread clinical application. Therefore, it is crucial to investigate the mechanisms involved in exosome formation, sorting, and content packaging, as well as their role in the dissemination

of abnormal proteins associated with NDs. Additionally, new methods for exosome modification, new drug action targets, and the identification of novel therapeutic pathways are urgently required.

Schematic diagram of therapeutic strategies for exosomes in neurodegenerative diseases. Exosomes inhibit M1 microglial activation and increase dendritic spine density in Alzheimer's disease models. The level of substantia nigra in the brain of Parkinson's disease rats can be increased after exosome therapy. Exosomes can effectively reduce abnormal aggregation of huntington protein in Huntington's disease mice. Exosomes significantly reduce demyelination in multiple sclerosis mouse models.

Exosome Modification

Upon entering the body, natural exosomes can be readily taken up by non-target cells or interact with cell membranes (de Abreu et al., 2020). This lack of specific recognition of target cells may lead to potential side effects and toxicity when interacting with non-target cells. To enhance their targeting precision, various strategies for modifying exosomes have been developed, including genetic engineering, chemical modifications (both covalent and non-covalent), and nanomaterial modifications (Lu et al., 2023). Bellavia et al. (2017) analyzed the size and morphology of exosomes and concluded that their modification did not alter their size or molecular characteristics. Surface-modified exosomes can cross the BBB more effectively than natural exosomes, making them suitable targeted drug-delivery carriers for site-specific drug administration. The ability of exosomes to deliver drugs to specific sites makes them promising tools for the treatment of NDs. Surface-modified exosomes are expected to find applications in *in vivo* imaging and tracking, highlighting their potential in drug-delivery research.

Genetic engineering

Genetic engineering can be used to modify exosomes and holds significant promise for enhancing the targeting capabilities and therapeutic potential of these vesicles. Transmembrane proteins on the surface of exosomes can be fused with ligands, thereby improving the specificity of exosome targeting (Mentkowski et al., 2018). This process involves fusing ligands or targeted peptides with transmembrane proteins expressed on the surface of exosomes. Exosome-producing cells are transfected with plasmids that encode these fusion proteins, allowing for the secretion of exosomes with targeted ligands. For instance, Li et al. (2019b) successfully improved the loading efficiency of a drug by developing CD9 HuR-functionalized exosomes through the transfection of HEK293T cells with a plasmid encoding CD9 HuR. Similarly, Alvarez-Erviti et al. (2011) fused rabies virus glycoproteins (RVG) with Lamp2b and transfected exosome-producing dendritic cells with siRNA to produce exosomes that deliver siRNA to the mouse brain, which improved outcomes in AD. In research by Kojima et al. (2018), RVG-Lamp2b-transfected exosome-producing HEK293T cells efficiently delivered mRNA while reducing the neurotoxicity and neuroinflammation associated with PD. Additionally, a study demonstrated that BMSC exosomes transfected with CXC motif chemokine receptor type 4 promoted microvascular endothelial cell proliferation after ischemic stroke, thus restoring vascular function and nerve repair (Li et al., 2020c).

The fusion of proteins with Lamp2b on the surface of exosomes has also been achieved through genetic engineering for targeted delivery. For example, Kim et al. (2018) co-transfected HEK293 cells with a plasmid encoding a heart-specific peptide linked to Lamp2b, resulting in the production of heart-targeting exosomes. This genetic modification of exosome membranes not only enhanced their targeting precision but also facilitated the loading of therapeutic molecules onto the exosomes. In another study, the Fe65 protein was used to modify HT22 hippocampal neuron-derived exosomes, which were then loaded with autophagy inducers to achieve their targeted delivery to amyloid precursor proteins, ultimately inducing neuronal autophagy and improving cognitive dysfunction in AD mice (Iyaswamy et al., 2023). A more efficient exosome, M2pep-ADSC-Exo, was developed through genetic engineering to specifically target M2 microglia, providing a novel method for the treatment of stroke (Wang et al., 2024).

While genetic engineering offers tremendous potential in exosome technology, such as rapid production, large volumes, and high specificity of modification, there are also several challenges associated with this approach. These include the high production costs, technical difficulties, challenges with distinguishing exosomes in biological fluids, and the possibility of errors in the expression process of high-molecular-weight ligands.

Chemical modification

Chemical modification enhances the targeting and therapeutic potential of exosomes through various methods, such as the attachment of natural or synthetic ligand receptors to their surfaces. These chemical modifications can be categorized as covalent and non-covalent modifications.

Covalent modifications

Covalent modifications play a crucial role in fixing targeted moieties onto exosomes (Rayamajhi and Aryal, 2020). For instance, covalently modified amine and carboxyphospholipids on the surface of extracellular vesicles can rapidly form chemical bonds with various functional groups. Among the techniques used for covalent modification, click chemistry is the most widely employed (An et al., 2019). This method involves a specific type of chemical reaction catalyzed by copper, in which alkyne and azide functional groups undergo a cycloaddition reaction. Importantly, this coupling reaction does not hinder the ability of exosomes to enter target cells (Smyth et al., 2014). Click chemistry is also highly efficient and can be performed in both organic solvents and aqueous buffers.

Jia et al. (2018) ingeniously modified exosomes using superparamagnetic iron oxide nanoparticles (SPIONs) and curcumin (Cur). They used click chemistry to couple the exosome membrane with a neuropilin-1-targeted peptide (RGE), resulting in glioma exosomes referred to as RGE-Exo-SPION/Cur. Remarkably, RGE-Exo were able to effectively traverse the BBB and persist in the tumor area for an extended period, indicating their potential for precise imaging and targeted therapeutic interventions (Jia et al., 2018). A prior study screened for optimal metabolic precursors for exosome labeling using bio-orthogonal click chemistry, leading to the development of resveratrol-containing macrophage exosomes (RSV&Exo) for treating MS. The intranasal administration of RSV&Exo significantly inhibited the inflammatory response in mouse models of MS (Zheng et al., 2023).

Metabolic engineering of exosomal blasts is another strategy for preparing surface-modified exosomes using click chemistry. This method introduces synthetically modified amino acids and lipids into the exosome culture medium, thereby decorating the exosome surfaces with modified proteins or lipids. The target components are then fused through click chemistry, resulting in surface-modified exosomes with tailored functionalities (Armstrong et al., 2017). One study demonstrated that chemically modified exosomes functionalized with heterogeneous bilayer scaffolds effectively repaired bone defects and muscle injuries (Shue et al., 2024). However, the non-specific binding of targeted molecules to exosome surfaces limits the use of click chemistry for exosome surface modification.

In conclusion, covalent modifications, particularly those added via click chemistry, offer a potent tool for enhancing the functionality of exosomes. These modifications enable the precise adherence of molecules onto exosome surfaces, thus enhancing their targeting capabilities and therapeutic potential for various applications. It should be noted that click-chemical-targeting molecules bind non-specifically to exosome surfaces, while metabolic engineering provides a more focused method.

Non-covalent modifications

Non-covalent techniques have also gained attention in exosome surface modification research. These techniques encompass a variety of strategies, including polyvalent electrostatic and hydrophobic interactions, ligand-receptor binding, aptamer-based modifications, and the incorporation of CP05-anchored peptides.

Polyvalent electrostatic interactions occur when negatively charged biofilms are targeted by positively charged regions of exosomes. Liposomes, which typically have positively charged groups, are commonly used for these electrostatic interactions (Tamura et al., 2017). In contrast, ligand-receptor interactions involve targeting ligands that bind to native receptors on the surface of exosomes. For instance, the RVG peptide was bound to MSC-exosomes through a DOPE-NHS linker, significantly enhancing the exosomes' ability to target the brains of transgenic APP/PS1 mice. This approach improved the cognitive ability, reduced plaque deposition and A β levels, and normalized the inflammatory cytokine levels of mice with AD (Cui et al., 2019). In a noteworthy study, Wang et al. (2017) demonstrated the successful integration of receptors onto exosome surfaces. Their process involved two steps: biotin was first attached to human umbilical vein endothelial cell-derived exosomes, and this was followed by a second attachment through a biotin-biotin interaction.

Hydrophobic interactions between liposomes and exosomes represent another strategy for surface modification. This method involves functionalizing liposome membranes with peptides, antibodies, or polyethylene glycol, followed by the fusion of exosomes and liposomal membranes using freeze-thaw methods. For example, Lee et al. (2016a) successfully fused liposomal and exosome membranes through a freeze-thaw technique; the resulting fusion products did not impair the membrane protein functions of the exosomes, which exhibited effective targeting capabilities. Additionally, Rui et al. (2024) successfully constructed liver-targeting transmembrane peptide-Exo-M1-8 through co-incubation, and effectively applied these in the targeted therapy of hepatocellular carcinoma.

Nucleic acid-based aptamers have also been developed for exosome surface modifications. Wan et al. (2017) reported that the LZH8 aptamer, which contains amplified nucleotide P, exhibited a strong binding affinity for hepatoma cell exosomes. The CP05 peptide offers a simpler route to targeted exosome surface expression. This peptide has a robust affinity for the second extracellular loop of CD63, enabling it to serve as a bridge between the targeted fraction and the exosome surface. Notably, the anchoring modification involving CP05 does not compromise the size, morphology, or *in vivo* distribution of exosomes (Kooijmans et al., 2016). Furthermore, the fusion of CP05-modified exosomes with muscle-targeting and RVG peptides can enhance their targeted delivery to surrounding muscles and the brain (Alvarez-Erviti et al., 2011; Gao et al., 2014).

In summary, non-covalent modification strategies offer versatile and effective means to modify exosome surfaces. These techniques encompass a diverse array of interactions, including electrostatic, receptor-based, hydrophobic, and aptamer-mediated interactions, providing researchers with a rich toolkit to engineer exosomes for precise targeting and therapeutic applications.

Nanomaterial modification

Metal-organic frameworks manifesting as intricate structural topologies in one-, two-, or three-dimensional formats and featuring 3D porous arrangements can be excellent exosome carriers. It has been postulated that the fusion of exosomes with such nanomaterials could be utilized in medical applications. Nanomaterial modifications can prevent exosomes from being absorbed by immune cells and enable their detection through diverse means, such as light, ultrasonic, and magnetic signals, for diagnostic purposes. Due to their hydrophilic core, exosomes face limitations in the swift loading of hydrophobic drugs. However, following modification with nanomaterials containing hydrophobic pores, exosomes can be loaded with hydrophobic drugs and rapidly delivered to cancer cells (Schindler et al., 2019).

Yang et al. (2020b) developed 3D-cultured exosomes that reduced A β production and improved memory and cognitive deficits in AD mice by promoting the upregulation of ADAM10 levels, the major component of α -secretase. Moreover, Liu et al. (2020) devised an exosome nanoscavenger for scavenging α -synuclein aggregates and reducing their cytotoxicity in PD neurons. Wang et al. (2022c) constructed an exosome nanotherapy platform to treat PD through multistep targeting and multistage intervention methods. Gold nanoparticles were integrated into exosomal phospholipid

membranes, to generate effective functioning peroxidase-like nanoenzymes (Di et al., 2020). Thus, exosome modification eliminates the need for post-labeling antibody detection. Zhuang et al. (2020) confirmed that the ability of exosomes to target tumors was significantly enhanced following their modification with superparamagnetic iron oxide nanoparticles. The resultant nano-drug delivery system exhibited a high drug loading rate, culminating in improved therapeutic effects, dose reduction, and minimal potential side effects. Furthermore, Li et al. (2022a) administered multifunctional nanomaterial-modified exosomes to a rat model with skull defects, resulting in the regulation of bone immune metabolism and the promotion of bone healing. In one study, functional exosomes containing superparamagnetic iron oxide nanoparticles were demonstrated to provide targeted delivery in the brains of mice. Based on this, these exosomes may be useful for the treatment of brain diseases (Wang et al., 2022b). The utilization of nanomaterials to engineer exosomes offers novel avenues for enhancing the precision and efficiency of cell-targeted therapy and is expected to expand their applications in the biomedical field.

Selection of exosome-modification strategies

Each of the above modification methods has advantages and disadvantages. Genetic engineering is more accurate for obtaining the desired product, but it is expensive and technically challenging. Chemical modification can be used to effectively control the surface functional structure of exosomes; however, the procedure is complicated, and protein function on the exosome membrane may be compromised during modification. Nanomaterial modification offers simplicity and speed, but it should be noted that the accumulation of nanomaterials can be potentially toxic, and the loading efficiency varies. These modification strategies present attractive prospects for enhancing the targeted delivery and therapeutic capabilities of exosomes. In the development of ND treatments, exosome-modification methods must be selected while considering several factors: the cell origin of the exosomes, the drug loaded, and the type of ND to be treated. Additionally, confirmation of the morphology of exosomes and characterization of their surface composition are essential after their modification.

In summary, exosomes can overcome the limitations of insufficient drug targeting through modifications, significantly improving the scope of their potential use in the therapeutic field. Modified exosomes are expected to be an effective means of treating ND.

Limitations

This review has several limitations. First, in addition to NDs, exosomes exert therapeutic effects on other conditions, such as cardiovascular disease (Ren et al., 2024), kidney disease (Yang et al., 2024), diabetes (Shi et al., 2024), and cancer (Bhatta et al., 2023; Caller et al., 2024; Li et al., 2024b), which are beyond the scope of this review. Second, exosomes can promote skin metabolism, slow aging (Wu et al., 2024a, b), and improve eye diseases (Yu et al., 2024), but these topics are not discussed here. Finally, this review does not summarize the clinical applications (Tian et al., 2023; Tan et al., 2024) or isolation methods (Yang et al., 2020a) of exosomes. In recent years, the incidence of NDs has increased annually; however, there have been no breakthroughs in their treatment. Meanwhile, engineered exosomes with higher targeting ability are increasingly used in the treatment of various diseases. Therefore, this article only reviewed the therapeutic potential of exosomes in NDs and the methods for enhancing exosome-targeted therapy.

Conclusion and Outlook

The incidence of NDs is increasing each year, and these conditions are difficult to diagnose and treat. Therefore, exploring the pathogenesis of NDs and searching for new early diagnostic markers have become focal points of research and attention in recent years. The advantages of using exosomes in clinical applications, such as providing a cell-free therapy that avoids the ethical issues associated with cell transplantation, possessing functions related to tissue damage repair and immune regulation, their increased stability *in vivo* compared to MSCs, their non-self-replicating nature

(reducing the potential risk of tumor formation), and their ability to penetrate the BBB, make them increasingly studied as carriers for targeted drug delivery. In this comprehensive review, we examined the therapeutic effects of exosomes in various NDs, including AD, PD, stroke, ALS, HD, MS, and epilepsy. We also summarized the surface modification methods used to enhance the targeting of exosomes, providing a valuable reference for the study of exosomes in NDs. However, the impact of exosomes extends beyond their critical role in ND treatment, and they hold promise as therapeutic tools in diverse medical domains. While enhancing the pathological landscape of several ailments, exosomes derived from the fluids or cells of pathological models have also become markers for diagnosing these conditions. The number of exosomes released, the cargo substances they carry, and their state are all inextricably linked to the condition of the originating cell.

The therapeutic effect of stem cell–derived exosomes in NDs is attributed to their ability to traverse the BBB, as well as their characteristics of stem cell differentiation and regeneration, low immunogenicity, and high survival rate following transplantation. However, stem cell–derived exosome therapy poses ethical and legal challenges. First, the origin of stem cells remains controversial. Currently, one of the main sources of stem cells is the embryo, which presents ethical and legal limitations. Second, the use and management of stem cells require the establishment of strict laws and regulations. Therefore, ethical and legal challenges must be considered when exploring the prospects of stem cell exosome therapy. Only within a reasonable ethical framework can stem cell exosomes be safely and sustainably developed for disease treatment.

As effective drug carriers, surface-modified exosomes enable precise drug delivery to target cells. Exosome therapy can alleviate the symptoms of NDs and has demonstrated efficacy in preclinical trials. However, several questions remain to be answered before it can be used in the clinic. First, exosomes come from a wide range of sources, but their surface glycoproteins vary with their cell source, and their contents are also diverse, which limits their clinical application and development. Second, obtaining numerous samples of high purity is challenging because the methods for exosome extraction and identification are complicated, necessitating the study of more convenient and efficient extraction techniques. Third, exosomes have a short half-life in the body and are quickly excreted, so further research is needed to understand the long-lasting effects of exosome therapy. Fourth, the optimal dose of exosomes for human therapy remains uncertain. Finally, the prognostic effects of exosome therapy are unclear, possibly due to the complexity of the mechanisms by which exosomes regulate disease and the lack of specific targets. Therefore, methods for the preparation, characterization, and optimization of exosomes should be improved in the future and their mechanisms of action and therapeutic potential in various diseases explored. Additionally, the side effects of exosomes require further investigation. Certain harmful proteins, nucleic acids, and other substances present in exosomes may exert adverse effects on the human body, including the hypothetical association between non-tumor disease treatment and increased cancer risk. Thus, the application range and safety of exosomes require further study and evaluation. The implementation of a new treatment is a gradual process. Exosomes serve as precision medicine carriers for NDs, and through genetic, chemical, and nanomaterial modifications, they can be innovative platforms for targeted delivery. A comprehensive understanding of the extraction, modification, and application of exosomes can elevate their importance in disease treatment research. In the future, exosomes may become effective interventions in the clinical treatment of diseases. With in-depth studies, they will continue to be integrated with regenerative medicine and other fields. Looking ahead, exosome therapy holds broad application prospects. Interdisciplinary collaborations and sustained investment in research are essential to realize this potential. In summary, exosomes may provide relief for individuals diagnosed with NDs. They transcend mere theoretical significance and exhibit substantial clinical research value in opening novel avenues for disease diagnosis and treatment.

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Footnotes

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Data Availability Statement

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